

Soccer Tournament Dashboard - Test Plan

1. Introduction

Scope: This test plan covers functional and non-functional aspects of the Soccer Tournament Dashboard, including core features and system behavior.

Core Features: Tournament selector, match schedule, standings, team profile, player stats, search, knockout bracket.

System Behavior: caching, background jobs, error handling, rate limiting, performance.

2. Roles and Responsibilities

Developer / QA / DevOps: Daniyal

3. Test Methodology

Unit Testing: pytest used for service logic and cache behavior.

Integration Testing: pytest + HTTPX for API routes and DB interaction.

Acceptance Testing: Playwright E2E flows based on user stories.

Regression Testing: all tests run via GitHub Actions on every PR.

Mutation Testing: mutmut used to validate test effectiveness.

Load Testing: k6 used to simulate traffic (normal, spike, stress).

4. CI/CD Workflow

Pull Requests trigger lint + unit + integration tests.

Merging to main deploys to staging.

Production deployment requires manual promotion.

Tests must pass before merge.

5. Test Levels

Unit: 10+ tests per core feature.

Integration: covers API + DB interactions.

Acceptance: validates user stories end-to-end.

Regression: automated on each commit.

Load: performance and stress scenarios.

6. Mutation Testing

Tool: mutmut.

Goal: high mutation kill rate (~85–100%).

Strategy: run after implementing service logic.

7. Load Testing

Tool: k6.

Scenarios: match fetch, standings fetch, search queries.

Spike test simulates high traffic during live matches.

8. Observability & Logging

Sentry for error tracking.

Render logs for runtime behavior.

GitHub Actions logs for background jobs.

Request logging middleware tracks latency and status.

9. Rate Limiting

Implemented using slowapi.

General: 100 requests/min/IP.

Search: 30 requests/min/IP.

Admin endpoints: strict limits.

Cache refresh cooldown prevents API abuse.

10. Terms and Acronyms

API: Application Programming Interface

AUT: Application Under Test

TTL: Time To Live

CI/CD: Continuous Integration / Continuous Deployment